SCAPMS Setup Guide: Get the App Running on Your Laptop

This guide takes you from a fresh Windows laptop to the clinic app running on screen. Follow the steps in order. The total time is about 1 to 2 hours, and most of that is waiting for downloads.

What you need before you start

- A Windows 10 or Windows 11 laptop (64-bit).
- At least 15 GB of free disk space. Check in File Explorer > This PC.
- A good internet connection. The Qt download is about 3 to 5 GB, so use Wi-Fi rather than mobile data if you can.
- An email address, to create a free Qt account.
- The file `SCAPMS_COS202.zip`.

Part 1: Install Qt and Qt Creator (one time only)

Qt is the toolkit the app is built with. Qt Creator is the program you use to open, build and run it.

Step 1. Create a free Qt account. Go to https://login.qt.io/register and sign up with your email. Open the confirmation email and click the link. (If the link does not open, skip this step and use the **Sign up** button inside the installer in Step 4.)

Step 2. Download the installer. Go to https://www.qt.io/development/download-qt-installer-oss and click the download button for **Windows x64**. You get a file named like `qt-online-installer-windows-x64-online.exe`.

If that page does not work, open https://download.qt.io/official_releases/online_installers/ and click `qt-online-installer-windows-x64-online.exe` to download it directly.

Step 3. Run the installer. Double-click the downloaded file. If Windows shows "Windows protected your PC", click **More info**, then **Run anyway**.

Step 4. Log in. Enter the email and password from Step 1 and click **Next**.

Step 5. Accept the open-source terms. Tick the box to accept the open-source obligations. Tick "I am an individual person not using Qt for any company". Click **Next**.

Step 6. Choose the folder. Leave the folder as `C:\Qt`. Choose **Custom installation** and click **Next**.

Step 7. Pick the parts to install. This is the most important step.

- Open **Qt**, then open the newest **Qt 6.x.x** in the list (for example Qt 6.8.3). Inside it, tick only **MinGW 64-bit** (the name includes a number like "MinGW 13.1.0 64-bit"). Leave the other compilers (MSVC, Android, WebAssembly) unticked. They are large and you do not need them.
- Open **Build Tools** (also called "Developer and Designer Tools"). Tick:
  - **MinGW 13.1.0 64-bit** (or whichever MinGW version matches the one you ticked above)
  - **CMake**
  - **Ninja**
- **Qt Creator** is ticked by default. Leave it ticked.

Click **Next**, accept the licence, then click **Install**.

Step 8. Wait. The download and install take 30 to 90 minutes. Keep the laptop plugged in and connected. Click **Finish** when it is done.

Part 2: Put the project on your laptop

Step 9. Make a projects folder. In File Explorer, open `C:\` and create a new folder named `Projects`. Avoid the Desktop, Documents or OneDrive folders, because OneDrive syncing can lock files while the app builds.

Step 10. Extract the zip. Right-click `SCAPMS_COS202.zip` > **Extract All...** and choose `C:\Projects` as the destination. You should now have the folder `C:\Projects\SCAPMS`, and inside it the folders `src`, `data`, `docs`, `tests` and the file `CMakeLists.txt`.

Part 3: Open, build and run the app

Step 11. Start Qt Creator. Open the Start menu, type **Qt Creator**, and open it.

Step 12. Open the project. Click **File > Open File or Project...**, go to `C:\Projects\SCAPMS`, select **CMakeLists.txt** and click **Open**.

Step 13. Choose the kit. Qt Creator shows a "Configure Project" page. Tick the kit named like **Desktop Qt 6.x.x MinGW 64-bit** and click **Configure Project**. Wait until the bottom of the window stops showing progress (about 30 seconds).

Step 14. Select the app as the thing to run. In the bottom-left corner there is a small computer icon. Click it and make sure **SCAPMS** is selected under "Run" (not `scapms_tests`).

Step 15. Build. Click the hammer icon at the bottom left, or press **Ctrl+B**. The first build takes 1 to 3 minutes. A green bar at the bottom right means it worked.

Step 16. Run. Click the green triangle at the bottom left, or press **Ctrl+R**. The SCAPMS window opens with 15 sample patients already loaded.

Part 4: Check that everything works

Click through each tab and try these:

1. **Dashboard:** numbers appear in the coloured cards.
2. **Patient Records:** click a patient row; the form on the right fills in. Change "Patient Type" and watch the form fields change. Type an age of 150 and click Add Patient; an error message appears.
3. **Appointments:** type a doctor's name, pick a future date and click Schedule Appointment.
4. **Waiting Queue:** add two patients, then click Call Next Patient. The first one added is served first.
5. **Search & Sort:** choose Merge Sort and Patient Name, click SORT. Then type part of a name and click Search.
6. **Reports:** switch between the four reports.
7. **Admin / Undo:** press Ctrl+Z to undo your last change.
8. **File > Save** (Ctrl+S) saves your changes. Close the app and run it again: the changes are still there.

Fresh dates. The sample appointments are dated early November 2026. To get sample data dated around today, use **Help > Load Demo Data**.

Run the automated tests (optional). Click the computer icon at the bottom left, choose **scapms_tests**, and press Ctrl+R. The "Application Output" panel should end with `76/76 checks passed`. Switch back to **SCAPMS** afterwards.

Take screenshots. Press **Windows + Shift + S**, drag over the app window, then paste into Paint or Word and save.

Part 5: If something goes wrong

Problem	Fix
No kit to choose in Step 13, or the kit has a red or yellow warning	Open `C:\Qt\MaintenanceTool.exe`, choose **Add or remove components**, and make sure Qt 6.x > MinGW 64-bit and Build Tools > MinGW, CMake and Ninja are ticked.
"Could not find a package configuration file provided by Qt6"	The wrong kit is selected. Click **Projects** on the left side bar, then **Manage Kits**, and choose the Desktop Qt 6 MinGW kit.
Build fails with red errors	Click **Issues** at the bottom of Qt Creator, copy the first red error and send it for help.
The app opens but shows "No data file found, loaded demo data"	Nothing is wrong. It means the program could not see the `data` folder, so it created sample data instead.
The app does not open when you press Run	Check the computer icon at the bottom left shows **SCAPMS**, not `scapms_tests`.
Windows Defender or antivirus blocks the program	Choose **Allow** or **Run anyway**. The app is built on your own laptop from the source code.
The download in Step 8 stops	Run `C:\Qt\MaintenanceTool.exe` again and choose **Update components**. It continues from where it stopped.

On a Mac instead of Windows

Use the same installer from Step 2 (choose the macOS version). In Step 7 tick Qt 6.x > **macOS**, plus **CMake** and **Ninja**. You also need Apple's compiler: open the Terminal app, type `xcode-select --install`, and press Enter. Then follow Parts 2 to 4 the same way. The kit in Step 13 is called **Desktop Qt 6.x.x clang 64-bit** or **macOS**.

What the main folders contain

- `src/model`: patient and appointment classes (PatientRecord is the abstract base class).
- `src/controller`: the logic, including the search algorithms (`SearchEngine.cpp`) and the sort algorithms (`Sorter.cpp`).
- `src/gui`: the windows and tabs.
- `src/utils`: saving and loading files, ID numbers and error types.
- `data/clinic_data.csv`: the saved patients and appointments.
- `docs/COS202_SCAPMS_Submission.pdf`: the document to submit.